

Hierarchical Clustering: Formulas and Numerical Example

Hierarchical clustering builds a hierarchy of clusters through iterative merging. The numerical example demonstrates how single linkage processes proximity in a bottom-up approach, with distance updates driving cluster formation.

1. Core Formulas

a) **Euclidean Distance** (measures similarity between points):

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

where x, y are data point vectors.

b) **Linkage Criteria** (define cluster proximity):

Single Linkage: Minimum distance between clusters

$$d(C_a, C_b) = \min_{x \in C_a, y \in C_b} d(x, y)$$

Complete Linkage: Maximum distance between clusters

$$d(C_a, C_b) = \max_{x \in C_a, y \in C_b} d(x, y)$$

Lance-Williams Algorithm (general update formula):

$$d(C_i \cup C_j, C_k) = \alpha_i d(C_i, C_k) + \alpha_j d(C_j, C_k) + \beta d(C_i, C_j) + \gamma |d(C_i, C_k) - d(C_j, C_k)|$$

For single linkage: $\alpha_i = \alpha_j = \frac{1}{2}, \beta = 0, \gamma = -\frac{1}{2}$

2. Numerical Example (Single Linkage)

Dataset: 6 points in 2D space

Point	X	Y
A	1	2
B	2	1
C	5	3
D	6	2
E	6	0
F	7	1

Step 1: Compute Initial Distance Matrix

Using Euclidean distance:

	A	B	C	D	E	F
A	0.00					
B	1.41	0.00				
C	4.47	4.24	0.00			
D	5.10	4.12	1.41	0.00		
E	5.39	4.47	3.16	2.00	0.00	
F	6.08	5.10	2.24	1.00	1.41	0.00

Step 2: Iterative Cluster Merging

1. **Merge 1:** Closest points D & F ($d=1.00$) → Cluster DF

Update distances (single linkage):

- $d(DF, C) = \min(d(D, C), d(F, C)) = \min(1.41, 2.24) = 1.41$
- $d(DF, E) = \min(d(D, E), d(F, E)) = \min(2.00, 1.41) = 1.41$
- $d(DF, A) = 5.10, d(DF, B) = 4.12$

2. **Merge 2:** Clusters DF & E ($d=1.41$) → Cluster DFE

Update distances:

- $d(DFE, C) = \min(d(DF, C), d(E, C)) = \min(1.41, 3.16) = 1.41$
- $d(DFE, A) = 5.10, d(DFE, B) = 4.12$

3. **Merge 3:** Clusters DFE & C ($d=1.41$) → Cluster DFEC

4. **Merge 4:** Points A & B ($d=1.41$) → Cluster AB

5. **Final Merge:** AB and DFEC ($d=4.12$)

Hierarchical Tree Structure:

$((D, F), E), C), (A, B)$

Dendrogram Visualization:

Height

```
3.0 |   DFEC
    |   /  \
    |  /    \
2.0 | /      \
    | /        \ AB
1.0 | DFE    /\
    | /  \   A B
0.0 | D   F
```

Points

3. Key Notes

1. **Algorithm Workflow:**

- Starts with each point as a singleton cluster.
- Iteratively merges the two closest clusters.
- Updates distances using linkage criteria after each merge.
- Ends with one all-inclusive cluster.

2. **Parameter Impact:**

- *Distance Metric:* Euclidean (used here) vs. Manhattan vs. Cosine.
- *Linkage Method:* Single linkage (chosen) vs. complete linkage vs. average linkage.

- Results vary significantly with these choices.

3. **Complexity:**

- Requires $O(n^2)$ memory for distance matrix.
- Suitable for datasets with $n < 10^4$ points.

4. **Implementation:**

Use Python's `scipy.cluster.hierarchy.linkage` with parameters:

```
from scipy.cluster.hierarchy import linkage
```

```
Z = linkage(points, method='single', metric='euclidean')
```

5. **Common Misconceptions:**

- Differs fundamentally from centroid-based methods (e.g., K-means) and density-based methods (e.g., DBSCAN).
- Generates nested clusters (dendrogram), not flat partitions.